

LECTURE 14

GROUP AND RING THEORY

We now show, via the use of vector spaces and modules, that every square matrix is similar to a certain matrix of block diagonal form.

Theorem 10.8. *Let V be a finite-dimensional vector space over a field F and let $T \in \text{Hom}_F(V, V)$. Then there exists a basis of V , as an $F[x]$ -module, with respect to which the matrix of T is*

$$\begin{pmatrix} B_1 & & & \\ & B_2 & & \\ & & \ddots & \\ & & & B_r \end{pmatrix},$$

where B_i is the companion matrix of a certain unique polynomial $f_i(x)$, for $i = 1, \dots, r$, such that $f_1(x) \mid f_2(x) \mid \dots \mid f_r(x)$.

Proof. Let V be an n -dimensional vector space over F . Then $\dim \text{Hom}_F(V, V) = n^2$, so if $0 \neq T \in \text{Hom}_F(V, V)$, the collection (T^i) , for $i = 0, 1, \dots, n^2$, must be linearly dependent over F . Let T^k be the first in this collection to be linearly dependent on the preceding ones. Thus,

$$T^r = a_0 I + a_1 T + \dots + a_{k-1} T^{k-1}$$

for some $a_i \in F$. Hence V as an $F[x]$ -module relative to T has the property that

$$\begin{aligned} 0 &= (a_0 I + a_1 T + \dots + a_{k-1} T^{k-1} - T^k)v \\ &= (a_0 + a_1 x + \dots + a_{k-1} x^{k-1} - x^k)v. \end{aligned}$$

Since $a_0 + a_1 x + \dots + a_{k-1} x^{k-1} - x^k \neq 0$, every element of the $F[x]$ -module V is a torsion element. As V is a finitely generated $F[x]$ -module, by Theorem 9.1, we have

$$V = V_1 \oplus \dots \oplus V_r,$$

where the $V_i [\cong F[x]/(f_i(x))]$ are cyclic $F[x]$ -modules, and $f_1(x) \mid f_2(x) \mid \dots \mid f_r(x)$. Also $(f_i(x))$ is the annihilator ideal of V_i in $F[x]$.

Since V_i is a T -cyclic space, the linear mapping T on V_i is represented by the companion matrix B_i of $f_i(x)$ with respect to a suitable basis $\mathcal{B}_i$ of V_i as shown in Theorem 10.6. So the linear transformation T on V is represented by the matrix A , which is the direct sum of the matrices $B_1, \dots, B_r$ with respect to the basis $\mathcal{B} = \cup_{i=1}^r \mathcal{B}_i$. $\square$

The matrix in the above theorem is called the *rational canonical form* of the matrix of T . The uniqueness of the decomposition of V subject to $f_1(x) \mid f_2(x) \mid \dots \mid f_r(x)$ shows that T has a unique representation by a matrix in a rational canonical form.

Example 10.9. Reduce the following matrix A to rational canonical form:

$$A = \begin{pmatrix} -3 & 2 & 0 \\ 1 & 0 & 1 \\ 1 & -3 & -2 \end{pmatrix}.$$

Solution. The invariant factors of $A - xI$ are obtained in Exercise Sheet 12, Question 1(b). These are $1, 1, (x+1)^2(x+3) = 3 + 7x + 5x^2 + x^3$. Therefore, the rational canonical form is

$$\begin{pmatrix} 0 & 0 & -3 \\ 1 & 0 & -7 \\ 0 & 1 & -5 \end{pmatrix}.$$

Example 10.10. The rational canonical form of the 6×6 matrix A whose invariant factors are $(x-3), (x-3)(x-1), (x-3)(x-1)^2$ is

$$\begin{pmatrix} 3 & & & & & \\ & \begin{pmatrix} 0 & -3 \\ 1 & 4 \end{pmatrix} & & & & \\ & & \begin{pmatrix} 0 & 0 & -3 \\ 1 & 0 & -7 \\ 0 & 1 & 5 \end{pmatrix} & & & \end{pmatrix}.$$

11. TENSOR PRODUCTS

Here we describe the concept of the tensor product of modules. Tensor products occur in many branches of mathematics and in physics.

This entire chapter is non-examinable for the oral exam.

11.1. Definition.

Definition 11.1. Let M, N , and P be abelian groups. A map $f : M \times N \rightarrow P$ is called *bi-additive* if

$$f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$$

and

$$f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$$

for all $m, m_1, m_2 \in M$ and $n, n_1, n_2 \in N$.

Example 11.2. The map $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by $(x, y) \mapsto xy$ is bi-additive, but if we had defined $(x, y) \mapsto x + y$, this would not be bi-additive, as $f(x_1 + x_2, y) = x_1 + x_2 + y \neq (x_1 + y) + (x_2 + y) = f(x_1, y) + f(x_2, y)$.

The above definition implies for a fixed $n \in N$, the map $m \mapsto f(m, n)$ from M to P and for a fixed $m \in M$ the map $n \mapsto f(m, n)$ from N to P are both homomorphisms. It follows that $f(m, 0) = 0 = f(0, n)$ and $f(m, -n) = -f(m, n) = f(-m, n)$ for all $m \in M$ and $n \in N$.

Definition 11.3. Let M be a right R -module and N be a left R -module. A map $f : M \times N \rightarrow P$ is called *balanced* if it is bi-additive and satisfies

$$f(mr, n) = f(m, rn)$$

for all $m \in M, n \in N$, and $r \in R$.

Example 11.4. The above example of a bi-additive map is clearly also balanced. See Exercise Sheet 15 for a non-example.

Throughout this chapter all rings have unity unless otherwise stated.

Definition 11.5. A *tensor product* of a right R -module M and a left R -module N is a pair (T, ϕ) , where T is an abelian group and ϕ is a balanced map from $M \times N$ to T satisfying the following condition: given any abelian group P and a balanced map $f : M \times N \rightarrow P$, there exists a unique homomorphism $\bar{f} : T \rightarrow P$ such that the diagram

$$\begin{array}{ccc} M \times N & & \\ \downarrow \phi & \searrow f & \\ T & \xrightarrow{\bar{f}} & P \end{array}$$

is commutative, that is, $\bar{f}\phi = f$.

Remark 11.6. It is easy to check (see Exercise Sheet 15) that if (T', ϕ') is another tensor product of M and N , then there exists an isomorphism $\alpha : T \rightarrow T'$ such that $\phi' = \alpha\phi$. Thus the tensor product is unique, up to isomorphism.

11.2. Existence and construction of a tensor product. We will now study an important construction which yields an abelian group when we are given both a right R -module M and a left R -module N . In this construction, the resulting group is the tensor product of the given modules M and N .

Let $A(M, N)$ denote the free abelian group (i.e. the free $\mathbb{Z}$ -module) with the set $M \times N$ as a basis. Let $B(M, N)$ denote the subgroup of $A(M, N)$ generated by the elements of the form

$$\begin{aligned} (m_1 + m_2, n) - (m_1, n) - (m_2, n), \\ (m, n_1 + n_2) - (m, n_1) - (m, n_2), \\ (mr, n) - (m, rn), \end{aligned}$$

for all $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$. Let $T(M, N) = A(M, N)/B(M, N)$ be the quotient group. We proceed to show that $T(M, N)$ is the tensor product of M and N . Let P be any abelian group and $f : M \times N \rightarrow P$ be a balanced map. We need to produce a balanced map $\phi : M \times N \rightarrow T(M, N)$ and a unique group homomorphism $\bar{f} : T(M, N) \rightarrow P$ such that the diagram

$$\begin{array}{ccc} M \times N & & \\ \downarrow \phi & \searrow f & \\ T(M, N) & \xrightarrow{\bar{f}} & P \end{array}$$

commutes. We choose ϕ to be the restriction of the canonical homomorphism

$$\pi : A(M, N) \rightarrow \frac{A(M, N)}{B(M, N)} = T(M, N)$$

to $M \times N$. Further, the map $f : M \times N \rightarrow P$ can be naturally extended to the free abelian group $A(M, N)$ with basis $M \times N$, yielding a homomorphism

$$\bar{f} : A(M, N) \rightarrow P.$$

However since f is a balanced map, the homomorphism $\bar{f}$ vanishes on the subgroup $B(M, N)$ and hence induces

$$\bar{f} : A(M, N)/B(M, N) \rightarrow P,$$

that is, from $T(M, N)$ to P ; where by abuse of notation we still write $\bar{f}$. It is clear that $\bar{f}\phi = f$. We note that $\bar{f}$ is unique. For, let $g : T(M, N) \rightarrow P$ be a homomorphism such that $g\phi = f$.

Then $g\phi(m, n) = \bar{f}\phi(m, n)$ for all $m \in M$ and $n \in N$. Since $\{\phi(m, n) \mid m \in M, n \in N\}$ generates $T(M, N)$, it follows that $g = \bar{f}$. This proves that $T(M, N) = A(M, N)/B(M, N)$ is the tensor product of M and N .